Pneumatic: sources, users, conventions

Item	Value
Engine bleed (high stages)	50-410 psi, 180-540 °C (idle → cruise)
Regulated bleed (reference)	44 psi (PRV), 200 °C (precooler loop)
Bleed ports	IP = HP stage 5 (normal, via IPCV); HP = HP stage 9 (idle, via HPV)
APU bleed	ground and flight up to 20 000 ft; feeds the LEFT side; valve opens at 95 % APU speed
Ground cart	lower LEFT fuselage connector, two-flap check valve; feeds the LEFT side
Hydraulic reservoirs	50 psi, from HP stage 8 of ENGINE 1 (deck drawing says 9: wrong)
MEL deferral	B 3 days, C 10 days, D 120 days
Numbering	odd = left (1), even = right (2)

Users: packs and pressurisation (ATA 21), engine start, wing anti-ice (from slat 3, piccolo duct) and engine anti-ice (ATA 30), hydraulic reservoirs (ATA 29), potable water (ATA 38), aft cargo heating (engine 1).

Engine bleed valves

Valve	Without pressure	Numbers	Commanded by
IPCV (stage 5)	check valve	opens when stage 5 > downstream	pressure difference
HPV (stage 9)	spring CLOSED	>8 psi opens; regulates 36 ± 4 psi; CLOSED if stage 5 > 36	pneumatic, slaved to PRV chamber C
PRV	spring CLOSED	≥8 psi fully open; regulates 44 ± 4 psi; fuse 450 °C	TLT sense line (solenoid: BMC / EEC)
OPV	spring OPEN	starts closing 74 psi, closed 90 psi (deck 73.8-89.9)	own pistons
FAV (fan air)	spring CLOSED	loop holds 200 ± 15 °C	TCT signal air
X-BLEED	electric, 2 motors	AUTO: same position as the APU bleed valve	motor 1 BMCs (AUTO), motor 2 manual
APU bleed valve	solenoid controlled	+ check valve against reverse flow	BMC via ECB

No bleed at all: HPV, PRV and FAV closed, only the OPV open. PRV closed ⇒ HPV closed, NOT the reverse (cruise: HPV closed while the PRV regulates). Valves can only reduce pressure, never add it.

Who regulates at each power (stage-5 pressure p5)

p5	IPCV	HPV	PRV	System gets
< 36 psi (idle)	closed (stage 9 higher)	regulates 36 ± 4	fully open	36 psi
36-44 psi	open	fully CLOSED	fully open	p5
> 44 psi (cruise)	open	closed	regulates 44 ± 4	44 psi
PRV fails high			BMC closes at 57 psi	0 (bleed off)
... and solenoid fails				OPV limits: 74-90 psi

Closing the PRV (and with it the HPV)

Manual: ENG BLEED pb OFF, ENG FIRE pb pushed (both energise the TLT solenoid).

BMC (electrical, TLT solenoid): > 57 psi after the PRV (Pr transmitter); > 257 °C after the precooler; leak detected; engine START sequence of that engine; APU BLEED selected (APU priority).

Pneumatic, inside the TLT: REVERSE flow (diaphragm: before-PRV vs after-precooler taps) → closes; over-temperature 235-245 °C → partial venting, PRV THROTTLES (does not shut).

PRV itself: upstream < 8 psi → spring closes it; thermal fuse melts at 450 °C.

Temperature ladder after the precooler

Temperature	Device	Action
200 ± 15 °C	TCT + FAV (pneumatic, continuous)	fan air through the precooler regulates
235-245 °C	TLT probe	bleeds the PRV sense line: PRV throttles back
257 °C	temperature sensor → BMC → TLT solenoid	PRV + HPV closed, ECAM caution
450 °C	PRV thermal fuse	vents the sense line: last resort

TCT: probe after the precooler (tube expands more than the inner rod). T < 200 °C: vent closed, chamber A full, needle valve closed, NO air to the FAV (FAV closed). T ≥ 200 °C: vent opens, needle valve opens, air to the FAV: proportional. Temperature is controlled ONLY by TCT, FAV, sense line and precooler (not by the BMC).

Precooler (in the pylon): hot bleed inside the tubes, fan air outside then overboard: heat exchange, no mixing.

Crossbleed, APU and engine start

X-BLEED AUTO opens (motor 1, either BMC) if: APU bleed valve OPEN + no ENG FIRE pb + no leak (a leak during a start sequence is tolerated until the sequence ends). APU valve closes → X-BLEED closes. OPEN/SHUT: motor 2, fixed, independent of the APU.

APU bleed valve opens if: APU BLEED pb ON, APU FIRE pb not pushed, no APU-duct leak (tolerated on ground until engines start), no APU shut-down signal, APU speed stabilised at 95 %. APU BLEED ON closes both engine PRVs.

Start with APU air: APU BLEED ON → APU valve and (AUTO) X-BLEED open → START valve → air starter → N2, N1. During its start the engine's PRV is held closed.

ENG FIRE pb: TLT solenoid → PRV + HPV of that engine closed, X-BLEED (AUTO) cannot open; hydraulic fire valve closes (ENG 1 Green, ENG 2 Yellow). APU FIRE pb: BMC refuses APU bleed. Two agents per engine, one for the APU.

Cockpit and computers

Overhead: ENG BLEED dark = normal, white OFF, amber FAULT; APU BLEED blue ON; X-BLEED AUTO/OPEN/SHUT. Dark cockpit: a light = abnormal or non-standard.

ECAM: class 1 RED (immediate action), class 2 AMBER (no immediate action), class 3 maintenance only (MCDU). BMC 1/2 identical; SDAC → FWC (alerts) + DMC (displays); CFDIU faults; EIU ← ECU; ECB (APU).

Hydraulic: three systems

System	Pumps	Notes
GREEN (main)	EDP on ENG 1	most users, biggest reservoir; fills ALL reservoirs (service panel)
YELLOW	EDP on ENG 2 + electric pump + hand pump	hand pump: cargo doors without power
BLUE (emergency)	electric pump + RAT	RAT also drives the CSM/G

All 3000 psi, fully independent (own reservoir, pumps, pipes). PTU = hydraulic motor + pump on one shaft: transfers POWER Green ↔ Yellow (no fluid) when dp > 500 psi, pb AUTO. PTU inhibited during engine start and cargo-door operation with the Yellow electric pump (and pb OFF).

Ailerons: Blue + Green. Elevators: left Green + Blue, right Blue + Yellow. Green users: gear and doors, nose-wheel steering, normal brakes, reverser ENG 1, flap/slat motors, THS, rudder, yaw damper, spoilers, aileron, elevator. Yellow: alternate brakes, reverser ENG 2, cargo doors. Blue: emergency generator, slats.

Hydraulic numbers (Green)

Item	Value
System pressure	3000 psi (EDP variable displacement, swashplate)
PTU threshold	dp > 500 psi
Reservoir air	50 psi (3.5 bar); LO AIR PRESS < 22 psi (Green, Yellow), < 30 psi (Blue)
Green reservoir	18 l max, 14 l normal, LOW LEVEL at 3 l (capacitive sensor)
Pump LO pressure	< 1740 psi
Manifold pressure switch	1450 psi (to ELAC, SEC, FAC, BSCU, ECAM): normal brakes lost below
Relief valve	opens 3500 psi, closes 3200 psi
Filters	HP and case drain clog at 87 psi dp; return filter 29 psi
Accumulator	N2 1885 psi at 20 °C, 1 l (steel + Kevlar)
Pipe band number	1 Green, 2 Blue, 3 Yellow; black dots = pressure line

$$pV = p_0 V_0 \;\Rightarrow\; V_{\text{fluid}} = V_0 \left(1 - \frac{p_{0,abs}}{p_{abs}}\right) = 1 \left(1 - \frac{1899.7}{3014.7}\right) = 0.37\,\text{l}$$

Priority valve: low pressure → cuts heavy users (gear, doors, steering, flap/slat motors) to keep the primary flight controls. Leak-measurement valve: normally OPEN, isolates the primary flight control users (tail: THS motor, rudder, yaw damper; each wing: aileron servo, spoilers 1 and 5) for internal-leak tests.

Fire valve: LH pylon, electric motor, closed by ENG 1 FIRE pb. ECAM HYD: pump in line green = delivering; LO amber < 1740 psi; system label amber when low; RAT automatic on electrical loss or RAT MAN ON (guarded).